

TEMPORAL LOCALIZATION OF SCORE ERROR AND THE ODDS FOR STOCHASTIC SAMPLING IN DIFFUSION MODELS

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ABSTRACT. The performance difference between probability-flow ODE sampling and stochastic reverse-time sampling depends not only on the magnitude of score approximation error but also on where that error is concentrated along the sampling trajectory. Motivated by this observation, we study a Gaussian model in which the score approximation error is parameterized by time-dependent functions. A logarithmic variance clock reduces the reverse dynamics to two scalar linear equations governing mean and variance mismatch; making it possible to have exact moment formulas for arbitrary time profiles.

For small variance error, the ODE–SDE comparison is controlled by explicit linear functionals of the error profile. We introduce an end-localization exponent κ through a profile proportional to $(V_0/V(t))^\kappa$. In the practically relevant long-horizon regime, a sharp transition occurs at the universal value $\kappa_c = 1$: shape errors with $\kappa < 1$ favor stochastic sampling, while sufficiently end-localized errors with $\kappa > 1$ favor deterministic sampling, independently of the stochasticity level $\gamma > 0$, up to leading order in the score error. Mean errors obey a different localization law, with critical exponent $\kappa_c = 0$. We further show that Gaussian-process priors on time-dependent score errors reduce, at leading order, to bivariate Gaussian orthant probabilities, yielding closed-form arcsine formulas for the probability that stochastic sampling improves over the ODE. These results suggest that the temporal geometry of score error, rather than its aggregate magnitude alone, is a natural variable for predicting when stochasticity helps.

1. INTRODUCTION

Sampling from a score-based diffusion model can be performed either by the probability-flow ODE or by a family of reverse-time SDEs with nonzero stochasticity. With an exact score, the corresponding continuous-time samplers share the same marginals. With an approximate score, however, the choice of stochasticity changes the generated distribution, and neither deterministic nor stochastic sampling is uniformly preferable.

Recent KL-divergence analysis makes the mechanism more precise: stochasticity can contract errors that have already accumulated in the generated distribution, while at the same time increasing the influence of the score error currently encountered by the sampler. Consequently, the effect of stochasticity depends on the *time localization* of score error, and not only on an integrated measure of its size; see [3]. This

suggests that a theoretical model in which score misspecification is represented by time-independent parameters may hide an important structural variable.

The goal of the present paper is therefore to make the time profile of score error part of the model from the outset. We work first in a one-dimensional Gaussian setting, not because time-independent Gaussian perturbations are assumed to be representative, but because the Gaussian setting remains analytically solvable even after the score-error parameters are promoted to arbitrary functions of time. The resulting formulas allow us to ask a more targeted question:

For which temporal profiles of score error does stochastic sampling improve over deterministic sampling, and is there a critical rate of end-localization separating the two regimes?

The answer is particularly simple after reparameterizing time by the logarithmic decrease of the forward variance. In this clock, arbitrary time-dependent affine score approximations lead to scalar linear equations for the mean and variance of the reverse process. For small variance-shape error, stochasticity acts as an exponential temporal filter. This leads to a general centroid criterion for infinitesimal stochasticity and, for an exponentially end-localized family, to a critical exponent $\kappa_c \simeq 1$. In the long-horizon limit the threshold is exactly 1 for every fixed $\gamma > 0$. Thus the same threshold separates ODE-favorable and SDE-favorable shape-error models across the whole stochastic sampler family.

A second contribution is probabilistic: we aim to obtain a probabilistic answer on the “odds” of stochastic sampling surpassing deterministic sampling, over a random family of error functions. At first sight this is infinite-dimensional. Near the exact score, however, the KL comparison depends only on a few linear projections of the error function. If the score-error profile is assigned a Gaussian-process prior, these projections are jointly Gaussian and the probability that stochasticity wins admits an arcsine formula. The infinite-dimensional prior therefore reduces to a low-dimensional covariance calculation.

Throughout most of the analysis the total log-variance horizon Λ is regarded as fixed. For commonly used diffusion scales it is already large, and the main phenomena below are much more sensitive to the localization of error than to moderate changes in Λ . We retain Λ in the exact formulas but emphasize the large- Λ limits, where the critical exponents become especially transparent.

2. TIME-DEPENDENT GAUSSIAN SCORE MODEL

2.1. Forward process and stochastic reverse family. Consider the one-dimensional variance-exploding forward diffusion

$$dX_t = g(t) dW_t, \quad X_0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad (1)$$

with

$$V(t) := \sigma_0^2 + \int_0^t g(s)^2 ds. \quad (2)$$

Then

$$p_t = \mathcal{N}(\mu_0, V(t)), \quad \nabla \log p_t(x) = -\frac{x - \mu_0}{V(t)}. \quad (3)$$

Let $\tau \in [0, T]$ denote reverse sampling time, so that the corresponding forward time is $t = T - \tau$. For a constant stochasticity parameter $\gamma \geq 0$, consider

$$dY_\tau = \frac{1+\gamma}{2} g(t)^2 s_\theta(Y_\tau, t) d\tau + \sqrt{\gamma} g(t) dB_\tau, \quad t = T - \tau. \quad (4)$$

The case $\gamma = 0$ is the probability-flow ODE, while $\gamma = 1$ is the usual reverse SDE for the VE forward process. More general γ interpolate and extrapolate between these cases.

We assume that the approximate score is affine in space but has arbitrary time-dependent coefficients:

$$s_\theta(x, t) = -\frac{x - \mu_\theta(t)}{\alpha_\theta(t)V(t)}, \quad \alpha_\theta(t) > 0. \quad (5)$$

The exact score corresponds to $\alpha_\theta(t) \equiv 1$ and $\mu_\theta(t) \equiv \mu_0$. The sampling prior is taken as

$$Y_0 \sim \mathcal{N}(\mu_T, \beta_T V(T)). \quad (6)$$

The score error is

$$\begin{aligned} \varepsilon_t(x) &:= s_\theta(x, t) - \nabla \log p_t(x) \\ &= \frac{\mu_\theta(t) - \mu_0}{\alpha_\theta(t)V(t)} + \left(1 - \frac{1}{\alpha_\theta(t)}\right) \frac{x - \mu_0}{V(t)}. \end{aligned} \quad (7)$$

Thus the model separates naturally into a *mean mode* and a *shape mode*. Their temporal profiles need not agree.

2.2. The logarithmic variance clock. Define

$$\ell := \log \frac{V(T)}{V(t)}, \quad 0 \leq \ell \leq \Lambda, \quad \Lambda := \log \frac{V(T)}{V(0)}. \quad (8)$$

Since $t = T - \tau$,

$$\frac{d\ell}{d\tau} = \frac{g(t)^2}{V(t)}. \quad (9)$$

We henceforth regard α_θ and μ_θ as functions of ℓ through the inverse relation $t = t(\ell)$, and abbreviate

$$\alpha(\ell) := \alpha_\theta(t(\ell)), \quad u(\ell) := \frac{\mu_\theta(t(\ell)) - \mu_0}{\sigma_0}. \quad (10)$$

Let

$$r_\gamma(\ell) := \frac{\mathbb{E}Y_\tau - \mu_0}{\sigma_0}, \quad v_\gamma(\ell) := \frac{\text{Var}(Y_\tau)}{V(t)}. \quad (11)$$

The following observation is the basic reason the time-dependent formulation remains tractable.

Proposition 2.1 (Exact moment dynamics for arbitrary time profiles). *For the reverse process (4)–(5), the law of Y_τ is Gaussian for every τ . In the log-variance clock, its standardized mean and normalized variance satisfy*

$$r'_\gamma(\ell) = -\frac{1+\gamma}{2\alpha(\ell)}(r_\gamma(\ell) - u(\ell)), \quad (12)$$

$$v'_\gamma(\ell) = \left(1 - \frac{1+\gamma}{\alpha(\ell)}\right)v_\gamma(\ell) + \gamma, \quad (13)$$

with initial conditions

$$r_\gamma(0) = r_T := \frac{\mu_T - \mu_0}{\sigma_0}, \quad v_\gamma(0) = \beta_T. \quad (14)$$

Proof. The drift in (4) is affine and the diffusion coefficient is deterministic, so Gaussianity is preserved. Writing $m_\tau = \mathbb{E}Y_\tau$ and $q_\tau = \text{Var}(Y_\tau)$, standard moment equations give

$$\begin{aligned} \frac{dm_\tau}{d\tau} &= -\frac{1+\gamma}{2} \frac{g(t)^2}{\alpha_\theta(t)V(t)}(m_\tau - \mu_\theta(t)), \\ \frac{dq_\tau}{d\tau} &= -(1+\gamma) \frac{g(t)^2}{\alpha_\theta(t)V(t)}q_\tau + \gamma g(t)^2. \end{aligned}$$

Using $d\ell/d\tau = g(t)^2/V(t)$ gives (12). For $v = q/V$, one additionally uses $dV/d\tau = -g(t)^2$, obtaining (13). $\square$

Define

$$A_\gamma(\ell) := \int_0^\ell \frac{1+\gamma}{2\alpha(s)} ds, \quad (15)$$

$$C_\gamma(\ell) := \int_0^\ell \left(1 - \frac{1+\gamma}{\alpha(s)}\right) ds. \quad (16)$$

Then the moment equations have the exact representations

$$r_\gamma(\Lambda) = e^{-A_\gamma(\Lambda)}r_T + \int_0^\Lambda e^{-[A_\gamma(\Lambda)-A_\gamma(s)]} \frac{1+\gamma}{2\alpha(s)} u(s) ds, \quad (17)$$

$$v_\gamma(\Lambda) = e^{C_\gamma(\Lambda)}\beta_T + \gamma \int_0^\Lambda e^{C_\gamma(\Lambda)-C_\gamma(s)} ds. \quad (18)$$

Thus promoting $(\alpha_\theta, \mu_\theta)$ from constants to functions does not remove analytical solvability; it replaces elementary powers by integrating-factor kernels.

Remark 2.1 (The time-independent model). If $\alpha(\ell) \equiv \alpha_\theta$ and $u(\ell) \equiv (\mu_\theta - \mu_0)/\sigma_0$, equations (17)–(18) reduce to the closed forms of the time-independent Gaussian model. In the present formulation this is merely the constant-profile special case.

2.3. Terminal KL divergences. At $\ell = \Lambda$ one has $V(t) = V(0) = \sigma_0^2$. Hence the generated distribution can be written

$$\tilde{p}_0^{(\gamma)} = \mathcal{N}(\mu_0 + \sigma_0 r_\gamma(\Lambda), \sigma_0^2 v_\gamma(\Lambda)). \quad (19)$$

The two KL directions are therefore

$$h_q(\gamma) := \text{KL}(\tilde{p}_0^{(\gamma)} \| p_0) = \frac{1}{2} (v_\gamma - \log v_\gamma - 1 + r_\gamma^2), \quad (20)$$

$$h_c(\gamma) := \text{KL}(p_0 \| \tilde{p}_0^{(\gamma)}) = \frac{1}{2} \left(v_\gamma^{-1} + \log v_\gamma - 1 + \frac{r_\gamma^2}{v_\gamma} \right), \quad (21)$$

where r_γ and v_γ on the right-hand sides mean their terminal values at Λ .

The principal comparison will be

$$h(\gamma) - h(0), \quad (22)$$

for fixed $\gamma > 0$, together with the local quantity $h'(0)$ when infinitesimal stochasticity is informative.

3. SHAPE ERROR AS A TEMPORAL FILTERING PROBLEM

We first isolate variance-shape misspecification by taking an exact prior and exact mean:

$$r_T = 0, \quad \beta_T = 1, \quad u \equiv 0. \quad (23)$$

The mean then remains exact, $r_\gamma \equiv 0$, for every γ .

Let the shape coefficient be a small arbitrary functional perturbation,

$$\alpha_\varepsilon(\ell) = \exp\{\varepsilon A(\ell)\}, \quad A \in L^\infty(0, \Lambda), \quad |\varepsilon| \ll 1. \quad (24)$$

The function A describes the temporal profile of the shape error, while ε controls its amplitude.

Proposition 3.1 (First variation of terminal variance). *For every fixed $\gamma \geq 0$,*

$$v_\gamma(\Lambda) = 1 + \varepsilon \mathcal{L}_\gamma[A] + O(\varepsilon^2), \quad (25)$$

where

$$\boxed{\mathcal{L}_\gamma[A] := (1 + \gamma) \int_0^\Lambda e^{-\gamma(\Lambda-s)} A(s) ds.} \quad (26)$$

Consequently, both KL directions $h_q(\gamma)$ and $h_c(\gamma)$ can be expressed in ε as

$$h(\gamma) = \frac{\varepsilon^2}{4} \mathcal{L}_\gamma[A]^2 + O(\varepsilon^3). \quad (27)$$

Proof. Since $\alpha_\varepsilon^{-1} = 1 - \varepsilon A + O(\varepsilon^2)$, equation (13) becomes

$$v'_\gamma = [-\gamma + (1 + \gamma)\varepsilon A + O(\varepsilon^2)]v_\gamma + \gamma.$$

Writing $v_\gamma = 1 + \varepsilon y_\gamma + O(\varepsilon^2)$ yields

$$y'_\gamma = -\gamma y_\gamma + (1 + \gamma)A, \quad y_\gamma(0) = 0,$$

whose terminal solution is (26). Finally, both Gaussian KL directions satisfy

$$\frac{1}{2}(v - \log v - 1) = \frac{1}{4}(v - 1)^2 + O((v - 1)^3),$$

and likewise for $\frac{1}{2}(v^{-1} + \log v - 1)$. $\square$

Remark 3.1 (Stochasticity as a temporal filter). The ODE sees the unweighted functional

$$\mathcal{L}_0[A] = \int_0^\Lambda A(s) ds, \quad (28)$$

whereas $\gamma > 0$ replaces it by an exponentially weighted functional. If $d = \Lambda - s$ denotes log-variance distance from the data end, then the kernel is

$$K_\gamma(d) = (1 + \gamma)e^{-\gamma d}. \quad (29)$$

Thus stochasticity suppresses errors encountered far from the end of sampling and gives comparatively greater weight to errors localized near the data scale.

Corollary 3.1 (Leading-order finite- γ comparison). *If $\mathcal{L}_\gamma[A]^2 \neq \mathcal{L}_0[A]^2$, then for sufficiently small nonzero ε ,*

$$h(\gamma) < h(0) \iff |\mathcal{L}_\gamma[A]| < |\mathcal{L}_0[A]|, \quad (30)$$

for both KL directions.

3.1. Infinitesimal stochasticity and a localization-centroid criterion. Differentiating (26) at $\gamma = 0$ gives

$$\mathcal{L}_0[A] = \int_0^\Lambda A(s) ds, \quad (31)$$

$$\left. \frac{d}{d\gamma} \mathcal{L}_\gamma[A] \right|_{\gamma=0} = \int_0^\Lambda (s + 1 - \Lambda) A(s) ds. \quad (32)$$

Hence

$$h'(0) = \frac{\varepsilon^2}{2} \left(\int_0^\Lambda A(s) ds \right) \left(\int_0^\Lambda (s + 1 - \Lambda) A(s) ds \right) + O(\varepsilon^3). \quad (33)$$

For a nonnegative profile $A \not\equiv 0$, define its mean distance from the data end Λ , in log-variance time, by

$$\bar{d}_A := \frac{\int_0^\Lambda (\Lambda - s) A(s) ds}{\int_0^\Lambda A(s) ds}. \quad (34)$$

Theorem 3.1 (Shape-error localization criterion). *Assume (23) and (24), with $A \geq 0$ and $A \not\equiv 0$. Then, to leading order as $\varepsilon \rightarrow 0$,*

$$\boxed{h'(0) < 0 \iff \bar{d}_A > 1.} \quad (35)$$

Thus infinitesimal stochasticity is favorable when the center of mass of the shape error lies more than one unit of log-variance time before the data end, and unfavorable when the error is concentrated within the final unit.

Proof. For $A \geq 0$, the first factor in (33) is positive. The second factor equals

$$\int_0^\Lambda (1 - (\Lambda - s))A(s) ds = \left(\int_0^\Lambda A(s) ds \right) (1 - \bar{d}_A).$$

The claim follows. $\square$

Remark 3.2 (Recovery of the old e^{-2} threshold). The time-independent shape model corresponds to $A \equiv 1$. Then $\bar{d}_A = \Lambda/2$, so Theorem 3.1 gives stochastic preference iff $\Lambda > 2$, equivalently

$$\frac{V(0)}{V(T)} < e^{-2}. \quad (36)$$

Thus the previously observed e^{-2} threshold is the constant-profile member of the more general localization criterion (35); it need not be treated as the main axis of the theory.

4. A CRITICAL EXPONENT FOR END-LOCALIZED SHAPE ERROR

The previous theorem suggests parameterizing not the total noise horizon but the rate at which score error becomes concentrated near the low-noise end. We therefore introduce the family

$$A_\kappa(s) := e^{-\kappa(\Lambda-s)}, \quad \kappa \geq 0. \quad (37)$$

Since $\ell = \log(V(T)/V(t))$,

$$A_\kappa(\ell) = e^{-\kappa(\Lambda-\ell)} = \left(\frac{V(0)}{V(t)} \right)^\kappa. \quad (38)$$

The parameter κ is therefore a localization exponent in the physical noise scale: $\kappa = 0$ gives a time-independent shape mismatch, while increasing κ suppresses the mismatch at high noise and concentrates it progressively closer to $t = 0$.

For this family, Proposition 3.1 gives

$$\mathcal{L}_0[A_\kappa] = \begin{cases} \frac{1 - e^{-\kappa\Lambda}}{\kappa}, & \kappa > 0, \\ \Lambda, & \kappa = 0, \end{cases} \quad (39)$$

$$\mathcal{L}_\gamma[A_\kappa] = (1 + \gamma) \frac{1 - e^{-(\gamma+\kappa)\Lambda}}{\gamma + \kappa}, \quad \gamma > 0. \quad (40)$$

Proposition 4.1 (Finite-horizon critical exponent). *Fix $\gamma > 0$. Suppose*

$$\frac{(1 + \gamma)(1 - e^{-\gamma\Lambda})}{\gamma\Lambda} < 1, \quad (41)$$

which holds in the usual large-noise-horizon regime. Then there is a unique critical exponent $\kappa_c(\gamma, \Lambda) > 0$ satisfying

$$(1 + \gamma) \frac{1 - e^{-(\gamma+\kappa_c)\Lambda}}{\gamma + \kappa_c} = \frac{1 - e^{-\kappa_c\Lambda}}{\kappa_c}. \quad (42)$$

For sufficiently small shape-error amplitude,

$$\kappa < \kappa_c(\gamma, \Lambda) \implies h(\gamma) < h(0), \quad \kappa > \kappa_c(\gamma, \Lambda) \implies h(\gamma) > h(0). \quad (43)$$

Proof. For $A_\kappa \geq 0$, it is enough to compare $\mathcal{L}_\gamma[A_\kappa]$ and $\mathcal{L}_0[A_\kappa]$. Their ratio can be written

$$R_\gamma(\kappa) = (1 + \gamma) \frac{\int_0^\Lambda e^{-(\gamma+\kappa)d} dd}{\int_0^\Lambda e^{-\kappa d} dd} = (1 + \gamma) \mathbb{E}_{\nu_\kappa}[e^{-\gamma d}], \quad (44)$$

where ν_κ is the probability measure on $[0, \Lambda]$ with density proportional to $e^{-\kappa d}$. As κ increases, ν_κ shifts strictly toward smaller d , while $e^{-\gamma d}$ is strictly decreasing in d . Hence $R_\gamma(\kappa)$ is strictly increasing. Condition (41) is exactly $R_\gamma(0) < 1$, while $R_\gamma(\kappa) \rightarrow 1 + \gamma > 1$ as $\kappa \rightarrow \infty$. The intermediate value theorem gives a unique crossing, and Corollary 3.1 gives (43). $\square$

The large-horizon limit removes almost all dependence on Λ and leads to the main critical-exponent statement.

Theorem 4.1 (Universal large-horizon shape threshold). *Fix $\gamma > 0$ and $\kappa > 0$. As $\Lambda \rightarrow \infty$,*

$$\mathcal{L}_0[A_\kappa] \longrightarrow \frac{1}{\kappa}, \quad \mathcal{L}_\gamma[A_\kappa] \longrightarrow \frac{1 + \gamma}{\gamma + \kappa}. \quad (45)$$

Consequently,

$$\boxed{\kappa < 1 \implies h(\gamma) < h(0), \quad \kappa > 1 \implies h(\gamma) > h(0)}, \quad (46)$$

for sufficiently small shape-error amplitude and sufficiently large Λ . The asymptotic critical exponent is

$$\boxed{\kappa_c = 1}, \quad (47)$$

independently of the fixed stochasticity level $\gamma > 0$.

Proof. The limits in (45) follow immediately from (39)–(40). Since both quantities are positive,

$$\begin{aligned} \mathcal{L}_\gamma[A_\kappa] < \mathcal{L}_0[A_\kappa] &\iff \frac{1 + \gamma}{\gamma + \kappa} < \frac{1}{\kappa} \\ &\iff \kappa(1 + \gamma) < \gamma + \kappa \\ &\iff \kappa < 1, \end{aligned}$$

using $\gamma > 0$. The opposite inequality holds for $\kappa > 1$. $\square$

Remark 4.1 (Practical horizon). For the commonly used EDM values $\sigma_{\text{data}} = 0.5$ and $\sigma_{\text{max}} = 80$, one has $V(0)/V(T) \approx 3.9 \times 10^{-5}$ and hence $\Lambda \approx 10.15$. At this horizon the finite- Λ correction in (42) is exponentially small: κ_c is already extremely close to 1 for standard values of γ . This is why we regard Λ mainly as a fixed scale parameter rather than a central experimental axis.

Remark 4.2 (Exponent of the actual score-error norm). For small ε , the shape component of (7) is approximately

$$\varepsilon A_\kappa(t) \frac{x - \mu_0}{V(t)}. \quad (48)$$

Its $L^2(p_t)$ norm scales as

$$\|\varepsilon_t^{\text{shape}}\|_{L^2(p_t)} \asymp |\varepsilon| V(t)^{-(\kappa+1/2)} \quad (49)$$

up to a fixed factor involving $V(0)^\kappa$. Thus $\kappa_c = 1$ corresponds to a critical RMS score-error growth proportional to $V(t)^{-3/2}$, or equivalently to the third inverse power of the standard deviation. The parameter κ should therefore be interpreted as an *additional* low-noise localization exponent beyond the geometric $V^{-1/2}$ factor already present in the shape score error.

5. RANDOM TIME PROFILES AND INFINITE-DIMENSIONAL “ODDS”

A natural objection to the time-dependent formulation is that $(\alpha_\theta(t), \mu_\theta(t))$ are now infinite-dimensional parameters. Near the exact score, however, the KL comparison only sees low-dimensional projections of these functions.

We first make this precise for shape error. Let A be a centered Gaussian process on $[0, \Lambda]$ with covariance kernel

$$K(s, t) := \mathbb{E}[A(s)A(t)], \quad (50)$$

and take $\alpha_\varepsilon = e^{\varepsilon A}$ with exact prior and exact mean. For each $\gamma \geq 0$, the quantity $\mathcal{L}_\gamma[A]$ in (26) is a centered Gaussian random variable.

Theorem 5.1 (Finite- γ arcsine formula for a Gaussian-process prior). *Fix $\gamma > 0$ and define*

$$Z_0 := \mathcal{L}_0[A], \quad Z_\gamma := \mathcal{L}_\gamma[A]. \quad (51)$$

Assume $Z_0 - Z_\gamma$ and $Z_0 + Z_\gamma$ are nondegenerate. Then, as $\varepsilon \rightarrow 0$,

$$\mathbb{P}(h(\gamma) < h(0)) = \frac{1}{2} + \frac{1}{\pi} \arcsin(\rho_\gamma) + o(1), \quad (52)$$

where

$$\rho_\gamma := \text{Corr}(Z_0 - Z_\gamma, Z_0 + Z_\gamma). \quad (53)$$

Equivalently, writing

$$V_0 := \text{Var}(Z_0), \quad V_\gamma := \text{Var}(Z_\gamma), \quad C_{0\gamma} := \text{Cov}(Z_0, Z_\gamma), \quad (54)$$

one has

$$\rho_\gamma = \frac{V_0 - V_\gamma}{\sqrt{(V_0 + V_\gamma - 2C_{0\gamma})(V_0 + V_\gamma + 2C_{0\gamma})}}. \quad (55)$$

In particular,

$$\mathbb{P}(h(\gamma) < h(0)) > \frac{1}{2} \iff \text{Var}(\mathcal{L}_\gamma[A]) < \text{Var}(\mathcal{L}_0[A]). \quad (56)$$

Proof. By Proposition 3.1,

$$h(\gamma) < h(0) \iff Z_\gamma^2 < Z_0^2$$

to leading order, except on a set whose probability vanishes as $\varepsilon \rightarrow 0$ under the stated nondegeneracy assumption. The event $Z_\gamma^2 < Z_0^2$ is equivalent to

$$(Z_0 - Z_\gamma)(Z_0 + Z_\gamma) > 0.$$

The two factors form a centered bivariate Gaussian vector, so the Gaussian orthant formula gives (52). Formula (55) follows from

$$\begin{aligned} \text{Cov}(Z_0 - Z_\gamma, Z_0 + Z_\gamma) &= V_0 - V_\gamma, \\ \text{Var}(Z_0 \mp Z_\gamma) &= V_0 + V_\gamma \mp 2C_{0\gamma}. \end{aligned}$$

The denominator in (55) is positive, proving (56). $\square$

The covariance terms in Theorem 5.1 are explicit quadratic forms of the GP kernel. Defining

$$w_\gamma(s) := (1 + \gamma)e^{-\gamma(\Lambda-s)}, \quad w_0(s) := 1, \quad (57)$$

one has

$$\text{Cov}(Z_\gamma, Z_\eta) = \int_0^\Lambda \int_0^\Lambda w_\gamma(s)K(s, t)w_\eta(t) ds dt. \quad (58)$$

Thus the infinite-dimensional prior enters the leading-order win probability only through three scalar quantities: V_0 , V_γ , and $C_{0\gamma}$.

5.1. Local arcsine formula. Define the second functional

$$D[A] := \int_0^\Lambda (s + 1 - \Lambda)A(s) ds. \quad (59)$$

Then (33) reads

$$h'(0) = \frac{\varepsilon^2}{2} \mathcal{L}_0[A]D[A] + O(\varepsilon^3). \quad (60)$$

Corollary 5.1 (Local GP odds). *If $(\mathcal{L}_0[A], D[A])$ is a nondegenerate centered bivariate Gaussian vector, then*

$$\mathbb{P}(h'(0) < 0) = \frac{1}{2} - \frac{1}{\pi} \arcsin(\text{Corr}(\mathcal{L}_0[A], D[A])) + o(1). \quad (61)$$

This is an infinite-dimensional counterpart of the scalar Gaussian orthant formulas: function space disappears after projection onto the two kernels relevant to the KL derivative.

5.2. Finite-dimensional basis priors. For computation and empirical calibration, one may replace a full Gaussian process by

$$A(s) = \sum_{j=1}^m \xi_j \phi_j(s), \quad \xi \sim \mathcal{N}(0, \Sigma). \quad (62)$$

Then

$$\mathcal{L}_\gamma[A] = k_\gamma^\top \xi, \quad (k_\gamma)_j = (1 + \gamma) \int_0^\Lambda e^{-\gamma(\Lambda-s)} \phi_j(s) ds, \quad (63)$$

and therefore

$$\text{Cov}(Z_\gamma, Z_\eta) = k_\gamma^\top \Sigma k_\eta. \quad (64)$$

This gives a practical route to fitting a prior over measured score-error profiles from independently trained networks.

5.3. Random localization exponents. An even lower-dimensional model keeps the functional nature of the error while randomizing its localization. Let

$$A(s) = Z A_\kappa(s), \quad (65)$$

where Z is a continuous random amplitude and κ is an independent random localization exponent. Since the leading KL depends quadratically on Z , the sign of the ODE–SDE comparison is controlled entirely by κ . Hence, in the large-horizon regime of Theorem 4.1,

$$\mathbb{P}(h(\gamma) < h(0)) \longrightarrow \mathbb{P}(\kappa < 1) \quad (66)$$

for every fixed $\gamma > 0$, up to the small-error approximation. This formulation makes the “odds” question directly a question about the distribution of temporal localization across trained models.

6. MEAN-ERROR PROFILES OBEY A DIFFERENT LOCALIZATION LAW

We next isolate mean misspecification by setting

$$\alpha \equiv 1, \quad \beta_T = 1, \quad r_T = 0, \quad (67)$$

and write

$$u(\ell) = \varepsilon U(\ell). \quad (68)$$

In this case $v_\gamma \equiv 1$ exactly for all γ , while the terminal standardized mean is

$$r_\gamma(\Lambda) = \varepsilon \mathcal{M}_\gamma[U], \quad (69)$$

where

$$\mathcal{M}_\gamma[U] := \frac{1 + \gamma}{2} \int_0^\Lambda e^{-\frac{1+\gamma}{2}(\Lambda-s)} U(s) ds. \quad (70)$$

Therefore both KL directions coincide exactly:

$$h_q(\gamma) = h_c(\gamma) = \frac{\varepsilon^2}{2} \mathcal{M}_\gamma[U]^2. \quad (71)$$

Differentiating at $\gamma = 0$,

$$\mathcal{M}_0[U] = \frac{1}{2} \int_0^\Lambda e^{-\frac{1}{2}(\Lambda-s)} U(s) ds, \quad (72)$$

$$\mathcal{M}'_0[U] = \frac{1}{4} \int_0^\Lambda e^{-\frac{1}{2}(\Lambda-s)} (2 - (\Lambda - s)) U(s) ds. \quad (73)$$

For $U \geq 0$, define the exponentially weighted mean distance from the data end

$$\bar{d}_U^{(m)} := \frac{\int_0^\Lambda (\Lambda - s) e^{-\frac{1}{2}(\Lambda-s)} U(s) ds}{\int_0^\Lambda e^{-\frac{1}{2}(\Lambda-s)} U(s) ds}. \quad (74)$$

Theorem 6.1 (Mean-error localization criterion). *Under (67)–(68), with $U \geq 0$ and $U \not\equiv 0$,*

$$\boxed{h'(0) < 0 \iff \bar{d}_U^{(m)} > 2.} \quad (75)$$

Thus the mean mode is already intrinsically weighted toward the end of sampling, even for the ODE, and stochasticity is favored only when the remaining weighted mean error lies sufficiently far from the data end.

Proof. Since (71) gives

$$h'(0) = \varepsilon^2 \mathcal{M}_0[U] \mathcal{M}'_0[U],$$

and $\mathcal{M}_0[U] > 0$, the sign is that of (73). Factoring out the positive weighted integral gives a factor $2 - \bar{d}_U^{(m)}$. $\square$

6.1. A critical exponent for the mean mode. Use the same end-localization family

$$U_\kappa(s) = e^{-\kappa(\Lambda-s)}. \quad (76)$$

Then

$$\mathcal{M}_\gamma[U_\kappa] = \frac{1 + \gamma}{2} \frac{1 - e^{-[\kappa + (1+\gamma)/2]\Lambda}}{\kappa + (1 + \gamma)/2}. \quad (77)$$

For $\kappa > -1/2$, the large-horizon limit is

$$\mathcal{M}_0[U_\kappa] \longrightarrow \frac{1}{2\kappa + 1}, \quad \mathcal{M}_\gamma[U_\kappa] \longrightarrow \frac{1 + \gamma}{2\kappa + 1 + \gamma}. \quad (78)$$

Hence, for every $\gamma > 0$,

$$\mathcal{M}_\gamma[U_\kappa] < \mathcal{M}_0[U_\kappa] \iff \kappa < 0. \quad (79)$$

Theorem 6.2 (Universal large-horizon mean threshold). *For the mean-error family (76), the asymptotic critical localization exponent is*

$$\boxed{\kappa_c^{(m)} = 0.} \quad (80)$$

Thus a mean-parameter error that grows toward the data end ($\kappa > 0$) favors the ODE, while a mean error that is larger earlier and decays toward the end ($\kappa < 0$) favors stochasticity. A time-independent mean mismatch ($\kappa = 0$) lies at the asymptotic

boundary; at finite horizon its sampler dependence is exponentially small when Λ is large.

Remark 6.1 (Why the shape and mean thresholds differ). The two score-error modes in (7) have different dynamical kernels. Shape error perturbs the covariance evolution, while mean error is itself propagated by the contracting reverse drift before it reaches the terminal distribution. It is therefore not generally meaningful to describe a trained model by a single scalar “score-error exponent” without specifying which spatial error mode is being measured. A useful empirical decomposition should distinguish at least translation-like and shape-like components, or work directly with the functional projections appearing in (26) and (70).

7. COMBINED SMALL MEAN AND SHAPE ERRORS

The preceding sections isolate two error modes. They can also be combined without losing the leading-order separation. Let

$$\alpha(\ell) = e^{\varepsilon_a A(\ell)}, \quad u(\ell) = \varepsilon_m U(\ell), \quad r_T = 0, \quad \beta_T = 1. \quad (81)$$

Then

$$v_\gamma(\Lambda) = 1 + \varepsilon_a \mathcal{L}_\gamma[A] + O(\varepsilon_a^2), \quad r_\gamma(\Lambda) = \varepsilon_m \mathcal{M}_\gamma[U] + O(\varepsilon_a \varepsilon_m). \quad (82)$$

Consequently, to total second order,

$$h_q(\gamma) = h_c(\gamma) = \frac{\varepsilon_a^2}{4} \mathcal{L}_\gamma[A]^2 + \frac{\varepsilon_m^2}{2} \mathcal{M}_\gamma[U]^2 + O(|\varepsilon_a|^3 + |\varepsilon_m|^3 + \varepsilon_a^2 |\varepsilon_m| + |\varepsilon_a| \varepsilon_m^2). \quad (83)$$

There is no second-order cross term in the Gaussian KL expansion because the leading mean and variance mismatches enter orthogonally.

Thus the ODE–SDE phase boundary is, to leading order,

$$\frac{\varepsilon_a^2}{4} (\mathcal{L}_\gamma[A]^2 - \mathcal{L}_0[A]^2) + \frac{\varepsilon_m^2}{2} (\mathcal{M}_\gamma[U]^2 - \mathcal{M}_0[U]^2) = 0. \quad (84)$$

Equation (84) makes explicit that “SDE-favorable” and “ODE-favorable” are properties of the temporal composition of score error. A model can contain an early shape error for which stochasticity is corrective and a late mean error for which stochasticity is harmful; the final verdict depends on their relative amplitudes and localization profiles.

For the exponential families A_{κ_a} and U_{κ_m} , the large-horizon functionals are

$$\mathcal{L}_0 = \frac{1}{\kappa_a}, \quad \mathcal{L}_\gamma = \frac{1 + \gamma}{\gamma + \kappa_a}, \quad \mathcal{M}_0 = \frac{1}{2\kappa_m + 1}, \quad \mathcal{M}_\gamma = \frac{1 + \gamma}{2\kappa_m + 1 + \gamma}, \quad (85)$$

whenever the relevant integrals converge. Substitution into (84) gives an explicit phase diagram in $(\kappa_a, \kappa_m, \varepsilon_m/\varepsilon_a, \gamma)$.

8. PRIOR MISMATCH

The score profile and the sampling prior represent distinct sources of error. Because the main purpose here is to understand temporal score-error localization, we have mostly used an exact prior above. Prior error can nevertheless be incorporated into the same projection framework.

For example, consider shape error with

$$\alpha = e^{\varepsilon A}, \quad \beta_T = e^{\varepsilon B}, \quad r_T = 0, \quad u = 0, \quad (86)$$

where B is a scalar perturbation. The first-order terminal variance is

$$v_\gamma(\Lambda) = 1 + \varepsilon \mathcal{L}_\gamma[A, B] + O(\varepsilon^2), \quad (87)$$

with

$$\boxed{\mathcal{L}_\gamma[A, B] := e^{-\gamma\Lambda}B + (1 + \gamma) \int_0^\Lambda e^{-\gamma(\Lambda-s)}A(s) ds.} \quad (88)$$

Thus prior-variance error is simply another mode concentrated at the initial boundary. If (A, B) is jointly Gaussian, Theorem 5.1 applies verbatim after replacing $\mathcal{L}_\gamma[A]$ by (88).

Similarly, mean-prior mismatch contributes the boundary term

$$e^{-\frac{1+\gamma}{2}\Lambda}r_T \quad (89)$$

to (17). For large Λ , this term is exponentially suppressed for every $\gamma \geq 0$, which is another reason to separate prior effects from the temporal score-error mechanism rather than combining them into a single scalar parameter model.

9. INTERPRETATION AND CONNECTION WITH KL ERROR EVOLUTION

The functional formulas above provide a solvable version of the qualitative mechanism identified in [3]. Stochasticity has two competing effects when the score is approximate: it increases contraction of discrepancies already present in the generated density, but it also changes how strongly the current score error enters the dynamics. The net effect depends on whether model error is predominantly an early perturbation that has time to be corrected, or a late perturbation injected close to the target distribution.

The kernel (29) makes this mechanism explicit in the shape mode. Relative to the ODE, stochastic sampling exponentially discounts error at log-variance distance d from the data end. The critical exponent of Section 4 arises from comparing this discount against the rate at which the model error itself grows toward $d = 0$. When the error profile grows more slowly than e^d in the corresponding reverse coordinate ($\kappa < 1$ in (37)), stochastic correction dominates. When the score error becomes concentrated faster than this critical rate, the SDE gives too much weight to the late inaccurate score and the ODE becomes preferable.

The mean mode shows that “time profile” should not be reduced to the scalar validation loss at each noise level without regard to error geometry. Different spatial components of the score error are propagated through different kernels. In higher dimensions one should therefore expect the relevant temporal statistic to involve projections of the score error onto dynamically meaningful modes, rather than a single norm alone.

10. EXPERIMENTAL PROGRAM

The theoretical predictions suggest experiments centered on the temporal profile of model error rather than on a sweep of the total noise horizon. We propose three increasingly realistic levels.

10.1. Controlled Gaussian experiment: the (κ, γ) phase diagram. Use the exactly solvable Gaussian model and inject the profiles

$$\log \alpha(\ell) = \varepsilon_a e^{-\kappa_a(\Lambda - \ell)}, \quad u(\ell) = \varepsilon_m e^{-\kappa_m(\Lambda - \ell)}. \quad (90)$$

For shape-only error, plot

$$h(\gamma) - h(0) \quad (91)$$

as a function of (κ_a, γ) for a fixed practical Λ . The predicted boundary is nearly vertical at $\kappa_a = 1$. For mean-only error the corresponding large-horizon boundary is $\kappa_m = 0$. The combined experiment can then vary the amplitude ratio $\varepsilon_m/\varepsilon_a$ and compare the empirical phase boundary with (84).

This experiment should be performed both with the exact moment formulas and by direct Euler–Maruyama simulation, separating the continuous-time mechanism from discretization effects.

10.2. Measure the error profile of trained networks. For a trained score network s_θ , estimate on validation data a noise-dependent error curve such as

$$E(t) := \mathbb{E}_{X_t \sim p_t} [\|s_\theta(X_t, t) - \nabla \log p_t(X_t)\|^2] \quad (92)$$

when the exact score is available, or an appropriate score-matching proxy otherwise. Rather than compressing $E(t)$ immediately into a single validation loss, fit its low-noise behavior to a localization model or project it onto kernels analogous to (26).

For synthetic data with known score, one can estimate separate mean-like and shape-like components directly. In higher dimensions, a first practical decomposition is into the component of score error parallel to $x - \mu_t$ and the orthogonal residual; more refined decompositions could use local Jacobian information or principal error modes.

10.3. Manipulate the time profile while controlling aggregate error. A particularly direct test of the mechanism is to train models with different time weightings in the score-matching objective. The goal is to produce networks with comparable integrated validation error but systematically different localization near the data end. If the present theory captures the relevant mechanism, models with slowly growing or early-concentrated shape error should benefit more from stochastic sampling than models whose error rises sharply at low noise.

Possible interventions include:

- changing the sampling distribution of training noise levels;
- multiplying the score-matching loss by controlled functions of log-noise time;
- early stopping or curriculum schedules designed to move error toward or away from the low-noise region;
- post-training controlled perturbations of the score in selected time windows.

The important design principle is to vary *where* the error occurs while keeping its global magnitude as comparable as possible.

10.4. An ensemble interpretation of the odds. Across independently trained networks, estimate either a basis prior (62) or a distribution of fitted localization exponents. The finite- γ probability predicted by Theorem 5.1 can then be compared with the observed paired win rate

$$\widehat{\mathbb{P}}(\mathcal{D}(\gamma) < \mathcal{D}(0)) \quad (93)$$

across training runs, where $\mathcal{D}$ is an appropriate sample-quality divergence or proxy. This preserves the original interpretation of a trained model as one draw from a population of imperfect scores, while allowing the dominant randomness to live in function space.

11. EXTENSIONS

11.1. Time-dependent stochasticity. The same log-variance formulation extends to a nonconstant stochasticity schedule $\gamma = \gamma(\ell)$. In the shape first variation, (26) is replaced by

$$\mathcal{L}_{\gamma(\cdot)}[A] = \int_0^\Lambda (1 + \gamma(s)) \exp \left\{ - \int_s^\Lambda \gamma(r) dr \right\} A(s) ds. \quad (94)$$

Thus choosing the stochasticity schedule is itself a problem of selecting a temporal filter for the score-error profile. This connects naturally with the optimal-control viewpoint developed for analytically solvable examples in [3].

11.2. Higher dimensions. For an isotropic n -dimensional Gaussian target with coordinatewise independent affine score errors, the preceding results add over coordinates. More generally, a time-dependent linear score

$$s_\theta(x, t) = -A_\theta(t)(x - \mu_\theta(t)) \quad (95)$$

leads to linear mean dynamics and a matrix Lyapunov equation for the covariance. The scalar kernels above are then replaced by matrix propagators. A useful next step is to identify whether analogues of the critical localization exponent survive after decomposition into eigendirections of the score Jacobian.

11.3. Beyond Gaussian data. The present calculations should not be interpreted as asserting that real score errors are affine. Their role is to expose, in a model where exact calculations remain possible, the dependence of ODE–SDE performance on temporal localization. For non-Gaussian data, the general KL evolution identities of [3] suggest analogous weighted temporal quantities, but the error geometry will depend on the evolving distribution and on the alignment between density mismatch and score error.

12. DISCUSSION

The time-dependent formulation changes the interpretation of the Gaussian model. The central random object is no longer a collection of static misspecification parameters but a score-error profile over the sampling trajectory. This seems closer to what is observed in trained networks, where model quality can vary by orders of magnitude across noise levels even when two networks have similar aggregate score-matching loss.

Three conclusions stand out.

First, arbitrary time dependence is analytically natural rather than technically prohibitive in the Gaussian model. The reverse moments remain governed by linear equations and are represented exactly by integrating factors.

Second, the small-error ODE–SDE comparison is low-dimensional even when the model prior is not. Shape error enters through the functionals $\mathcal{L}_\gamma$, mean error through $\mathcal{M}_\gamma$, and Gaussian functional priors produce explicit orthant probabilities. This suggests that the relevant statistical problem is to learn the covariance of a small number of temporal projections rather than a full probability law on score functions.

Third, the localization exponent gives a concrete phase-transition picture. For shape misspecification, an error profile proportional to $(V_0/V(t))^\kappa$ has critical exponent $\kappa_c = 1$ in the long-horizon regime, independent of the fixed stochasticity level. Mean error has a different threshold, $\kappa_c = 0$. The distinction warns against treating the norm of score error as the sole predictor of sampling behavior; both temporal localization and spatial error mode matter.

A central empirical question is therefore whether trained diffusion models exhibit a reproducible distribution of localization exponents or, more generally, whether their measured error profiles have projections that predict the sign and size of the ODE–SDE performance gap. If so, the theory would turn the qualitative statement that “the time profile matters” into a quantitative criterion for choosing stochasticity from properties of the trained score network.

13. OPEN PROBLEMS

The following directions appear especially natural.

- (i) **Nonperturbative shape error.** Determine whether the critical exponent $\kappa_c \approx 1$ persists beyond the small- ε expansion for $\alpha(\ell) = \exp\{\varepsilon A_\kappa(\ell)\}$.
- (ii) **Optimal stochasticity schedule.** Given an estimated error profile A , minimize $|\mathcal{L}_{\gamma(\cdot)}[A]|$ or the corresponding nonlinear terminal KL over admissible schedules $\gamma(\ell)$.
- (iii) **Profile identifiability from training diagnostics.** Determine which validation quantities provide stable estimates of the functionals $\mathcal{L}_\gamma$ and $\mathcal{M}_\gamma$ without requiring access to the exact score.
- (iv) **Multimode error geometry.** Extend the mean/shape decomposition to local eigendirections of the score Jacobian and investigate whether each mode possesses its own localization threshold.
- (v) **Connection with KL bounds.** Relate the exact Gaussian filtering kernels to the time weights appearing in general relative-entropy inequalities, with the aim of deriving sufficient profile conditions for stochastic improvement beyond Gaussian models.

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